

Dino Arla

dino.arla@pln.co.id | dinoarla@gmail.com | dinoarla.com | he/him
orcid.org/0009-0005-7997-4653 | Google Scholar | linkedin.com/in/dino-arla

PROFILE & RESEARCH INTERESTS

Professional engineer and applied machine learning researcher at PLN, Indonesia's state electricity utility, with over eight years of industry-embedded experience developing and deploying analytical systems for national power distribution networks. My research focuses on machine learning for critical energy infrastructure: non-technical loss (NTL) mitigation, probabilistic modeling of customer behavior, cost-sensitive and optimization-driven model selection, and, more recently, sequential decision-making under uncertainty. I am interested in making these methods more reliable, interpretable, and operationally robust in high-consequence deployment environments.

EDUCATION

Sepuluh Nopember Institute of Technology (ITS), Surabaya, Indonesia

Aug 2025 – Feb 2026

BAN-PT institutional accreditation: Unggul (A)

Professional Engineer Program, (Engineer Degree - Ir.) — GPA 3.79/4.00

Sepuluh Nopember Institute of Technology (ITS), Surabaya, Indonesia

Aug 2023 – Aug 2025

BAN-PT institutional accreditation: Unggul (A)

Master of Technology Management (M.MT), Business Analytics — GPA 3.63/4.00 ("Very Satisfactory"), 36 credits

- Thesis: "Optimization of Electricity Theft Detection Based on Automated Machine Learning Through Multivariable and Cost-Sensitive Learning Approaches: A Case Study at PT PLN (Persero) Distribution Unit of West Java". Advisor: Dr. Achmad Choiruddin, S.Si., M.Sc. (grade A)
- Relevant coursework: Predictive Analytics & Machine Learning, Big Data Analytics, Decision Systems & Analysis, Business Statistics.
- Awardee, 2025 Interdisciplinary School of Management & Technology Research Grant Scheme

Tanjungpura University, Pontianak, Indonesia

Sep 2013 – Jul 2017

BAN-PT institutional accreditation: A

Bachelor of Engineering (S.T.), Control Engineering — GPA 3.45/4.00 (equivalent to UK Upper Second Class Honours)

- Final project: "AQORTA: An Integrated Air Quality and Noise Telemetry System with Solar Power System". Advisor: Prof. Dr. Eng. Ir. Ferry Hadary, S.T., M.Eng., IPM., CRA. (grade A)
- Awardee, Dean of the Faculty of Engineering Award for Fastest Graduate

PUBLICATIONS, BOOK & PATENT

- [1] Kinnarkar, R., Arief, M. M., Akhra, Y., & **Arla, D.** (2026). Comparing Socially-Equitable Renewable Energy Budget Allocation MDP Policies in Mature and Emerging Economies. *2026 International Conference on Industrial Engineering and Engineering Management (IEEM 2026)*. **Accepted; to appear**. Preprint: [arXiv:2607.10201](https://arxiv.org/abs/2607.10201)
- [2] Firdaus, R. W., **Arla, D.**, & Arief, M. M. (2026). Particle Swarm Optimization of Air Gap Distance for Externally Gapped Line Arrester (EGLA) on High Isokeraunic 70 kV Transmission Line Based on Field-Measured Lightning Data. *2026 International Conference on Technology and Policy in Energy and Electric Power (ICT-PEP)*. **Accepted; to appear**. DOI: pending (IEEE Xplore)
- [3] Haramaini, Q., **Arla, D.**, & Sidiq, M. H. (2026). Zonal Electricity Pricing Based on Substation Capacity: Leveraging Oversupply for Economic Equity and Industrial Competitiveness in Indonesia. *The 18th International Conference on Information Technology and Electrical Engineering 2026*. **Accepted; to appear**. DOI: pending
- [4] **Arla, D.**, & Choiruddin, A. (2025). Machine Learning-based Electricity Theft Detection Considering Customer Consumption Pattern and Geographical Condition. *2025 International Conference on Data Science and Its Applications (ICoDSA)*, 534–539. DOI: [10.1109/ICoDSA67155.2025.11157579](https://doi.org/10.1109/ICoDSA67155.2025.11157579)
- [5] **Arla, D.**, Sanjaya, C. W., Hidayah Sidiq, M., Haramaini, Q., & Fadliqaf, S. H. (2025). Eco En Liza (Utilization of Diesel Engine Coolant Waste Energy as a Source of Electricity for Micro Hydroelectric Power Plants). *2025 International Conference on Technology and Policy in Energy and Electric Power (ICT-PEP)*, 348–353. DOI: [10.1109/ICT-PEP67281.2025.11232224](https://doi.org/10.1109/ICT-PEP67281.2025.11232224)
- [6] Haramaini, Q., **Arla, D.**, & Sidiq, M. H. (2025). EnerBit (Energy to Bitcoin): A Strategy Model for Monetizing Idle Power Surplus into Indonesia National Foreign Exchange. *5th ASEAN International Conference on Energy and Environment (AICEE)*, Kuala Lumpur, 15 Oct 2025, extended abstract, p. 138. [Proceedings PDF](#)
- [7] Haramaini, Q., Setiawan, A., Sidiq, M. H., **Arla, D.**, & Garniwa, I. (2024). Exploring the Relationship between Solar Panel Adoption and Socio-Economic Factors: A Case Study of Urban Households in Developing Countries. *Journal of Electrical Systems*, 20(5s), 2737–2754. DOI: [10.52783/jes.2756](https://doi.org/10.52783/jes.2756)
- [8] Sidiq, M. H., **Arla, D.**, Fadliqaf, S. H., & Haramaini, Q. (2024). Anti Theft Power Smart Metering System. *2024 International Conference on Technology and Policy in Energy and Electric Power (ICT-PEP)*, 287–290. DOI: [10.1109/ICT-PEP63827.2024.10732892](https://doi.org/10.1109/ICT-PEP63827.2024.10732892)

- [9] Haramaini, Q., **Arla, D.**, & Garniwa, I. (2023). A Telemetry System for Real-Time Air Quality and Noise Monitoring in Urban Areas Based on IoT Technology. *Tuijin Jishu/Journal of Propulsion Technology*, 44(6), 446–455. [Journal page](#)
- [B] Hadary, F., Rezeki, S., Jati, D., **Arla, D.**, & Setyawan, W. (2023). *Inovasi Sistem Pengukuran Kualitas Udara Terintegrasi Solar Power System Berbasis Internet of Things (IoT)*. Penerbit Litnus (PT Literasi Nusantara Abadi Grup), 129 pp. ISBN 978-623-8227-97-6. [Publisher page](#)
- [P] Hadary, F., **Arla, D.**, Rezeki, S., Jati, D., & Rustamaji, R. M. (2023). AQORTA (Air Quality and Noise Real Time Data Acquisition) Sistem Telemetri Kualitas Udara dan Kebisingan Terintegrasi Solar Power System. Indonesian Patent Application No. P00202112421. [DGIP Official Gazette](#)

RESEARCH & PROFESSIONAL EXPERIENCE

Team Leader of Energy Transaction, PLN Distribution Unit of West Java, Bandung **Feb 2023 – Present**

- Lead a team that developed and deployed MAGNETO, an ML ranking system (XGBoost with cost-sensitive learning, trained on 4.6 million inspection records) that prioritizes electricity-theft (P2TL) inspections across 17 distribution units in West Java.
- Measured recovery of 14.1 GWh (IDR 22.1 billion, ≈ USD 1.3 million) of back-billed energy across West Java as of May 2025, per PLN inspection records.

Assistant Analyst, Electrical Network & Customer Data Mapping, PLN West Java, Bandung **Jan 2022 – Jan 2023**

- Developed KINEMATICS 4DX, a performance-monitoring and analytics system integrating spatial and operational data; approx. IDR 271 million in annual efficiency gains and cost savings.

Assistant Engineer, Transaction Meter Maintenance, PLN West Java, Bandung **Jan 2019 – Dec 2021**

- Designed and implemented TX-Cent, a centralized transaction-meter execution and monitoring system; approx. IDR 460 million in annual financial gains and operational savings.

Junior Engineer, Technology & Information, PLN Kapuas Power Plant Sector, Kubu Raya **Jul 2018 – Dec 2018**

- Created SIMBA Apps, a GUI-based fuel-oil information system for power-plant operations; estimated annual impact of IDR 488 billion in cost savings and efficiency improvements.

AWARDS & CREDENTIALS

- Best Paper and Best Presenter of ICoDSA 2025 (The 8th Internasional Conference on Data Science and Its Applications), Jakarta, **Jul 2025**
- 3rd Winner, Learning Innovation Knowledge Exhibition (LIKE) 2024, PLN Research and Development Centre, Grand Final, Jakarta, **Oct 2024**
- Golden Ticket Awardee, Innovation Contest 2024, PLN Research and Development Centre, Regional selection, Padang, **Aug 2024**
- 1st Winner, Innovation Contest 2024, PLN Distribution Unit of West Java, Bandung, **Jun 2024**
- 1st Winner, Innovation Contest 2022, PLN Distribution Unit of West Java, Bandung, **May 2022**

TEACHING & ACADEMIC SERVICE

- Speaker, Knowledge Hub “Innovating Future Energy System”, Enlit Asia 2026, **Sep 2026**
- Guest Lecture, “Applied Machine Learning for Business Value Creation: From Data Science to Real-World Strategy”, Faculty of Science and Data Analytics, Dept. of Statistics, ITS, **Nov 2025**
- Speaker, Knowledge Sharing Forum “Knowledge to Impact, Toward the Global 500”, PLN Learning Centre Pandaan, **Jun 2025**
- Instructor, Innovation Paper & Presentation, PLN Nusantara Power, Kapuas Power Generation Unit, **Feb 2025**
- Instructor, “MAGNETO: Machine Learning for Ranking System to Prioritize P2TL Target Operation”, PLN West Java, **Jun 2024**
- Co-Supervisor, Undergraduate Thesis: Assisted in supervising a bachelor student’s thesis project in the Department of Statistics ITS, focusing on Machine Learning, Anomaly Detection and Spatial Statistics applied to energy distribution data, Faculty of Science and Data Analytics, Dept. of Statistics, ITS, **Jun – Nov 2025**

SKILLS

- **Programming:** Python (scikit-learn, pandas, NumPy, matplotlib, seaborn), Julia, MATLAB, SQL
- **Machine learning:** PyCaret (AutoML), classification, anomaly detection, cost-sensitive learning, spatial statistics
- **Data & visualization:** Power BI, Tableau, Plotly, Google Colab, Jupyter, Oracle, ArcGIS
- **Tools:** Git, VS Code, Claude Code (AI-assisted development), Microsoft Excel (advanced)
- **Languages:** Indonesian (native), English (professional working proficiency)